



Module Code & Module Title

Level 5 – Network Operating Systems

Assessment Type

Logbook 9

Semester

2025/26 Autumn

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Assignment Due Date: Tuesday, December 23, 2025

Assignment Submission Date: Tuesday, December 23, 2025

Submitted To: Santosh Parajuli

Word Count (Where Required): 799

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1 Introduction

1.1 Aim

The main aim of this lab is to practice UNIX commands, including file manipulation using “cat”, searching files and searching text in files using “grep” command.

1.2 Objectives

- To create and navigate directory.
- Create file using cat command
- Using grep command to search and filter the text content.
- Create, use and remove the shortcut command.
- Use history and re-execute previous commands.

1.3 Required tools and concepts

Tools:

- Linux terminal (Debian).

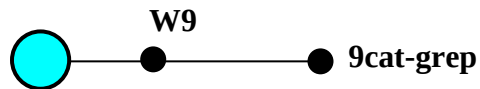
Concepts:

- Commands: mkdir, cd, grep , alias, unalias.
- Relative paths and absolute path.

1.4 Steps Required for Lab

Step-1: Create Directory Structure.

Created the directories as shown in the diagram using `mkdir -p`



Your
home
directory

```
manoj@DESKTOP-I4F9RFA:~$ mkdir -p w9/9cat-grep
```

Figure 1

```
manoj@DESKTOP-I4F9RFA:~$ tree
.
├── w9
│   └── 9cat-grep
3 directories, 0 files
```

Figure 2

Step:2 Changing to **9cat-grep** directory by one step using a relative pathname.

```
manoj@DESKTOP-I4F9RFA:~$ cd w9/9cat-grep
```

Figure 3

Step 4: Using the **cat** utility, creating two files

testa	testb
Kkkll	KKKKK
llmm	LLLLL
oo--oo	MMMMM
mmmdd	DDDDD
dddkk	

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ cat > testa
Kkkll
llmm
oo--oo
mmmdd
dddkk
```

Figure 4

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ cat > testb
KKKKK
LLLLL
MMMMM
DDDDD
```

Figure 5

Step 5: Use “grep” commands.

1. grep ll testa

The command finds all the line in testa file that contain the letter ll.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep ll testa
Kkkll
llmm
```

Figure 6

2. `grep -v ll testa`

The `grep` command finds all the line in `testa` that doesnot contain the letter `ll`.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep -v ll testa
oo-oo
mmmdd
dddkk
```

Figure 7

3. `grep -n ll testa`

The command finds the lines with `ll` and shows their line numbers.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep -n ll testa
1:Kkkll
2:llmm
```

Figure 8

4. `grep -l ll *`

The command list only filenames that contain `ll` in the current folder.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep -l ll *
testa
```

Figure 9

5. `grep -i ll *`

The command finds all the `ll` in all files ignoring capital or small letters (case insensitive).

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep -i ll *
testa:Kkkll
testa:llmm
testb:LLLLL
```

Figure 10

6. `grep -i LL *`

The command finds all the `LL` in all files ignoring capital or small letters (case insensitive).

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep -i LL *
testa:Kkkll
testa:llmm
testb:LLLLL
```

Figure 11

7. `grep -c ll *`

The command count how many lines in each file contain ll and shows the count.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep -c ll *
testa:2
testb:0
```

Figure 12

8. `grep '^K' testa testb`

The command find the lines start with letter K in both testa and testb.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep '^K' testa testb
testa:Kkkll
testb:KKKKK
```

Figure 13

9. `grep -n '^' testa`

The command shows all lines in testa with their line numbers.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ grep -n '^' testa
1:Kkkll
2:llmm
3:oo-oo
4:mmdd
5:ddkk
```

Figure 14

Step 6: Defining the **lsal** alias for **ls -al** command

Show that your system stores it giving the **alias** command (without arguments).

Use it in your home directory.


```
manoj@DESKTOP-I4F9RFA:~$ cd w9/9cat-grep
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ alias lsal='ls -al'
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ alias
alias ls='ls --color=auto'
alias lsal='ls -al'
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ |
```

Figure 15

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ lsal
total 16
drwxr-xr-x 2 manoj manoj 4096 Dec 21 02:35 .
drwxr-xr-x 3 manoj manoj 4096 Dec 21 02:32 ..
-rw-r--r-- 1 manoj manoj 29 Dec 21 02:35 testa
-rw-r--r-- 1 manoj manoj 24 Dec 21 02:35 testb
```

Figure 16

Step 7: Removing the alias. And also showing that system does not store it.

```
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ unalias lsal
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ lsal
-bash: lsal: command not found
manoj@DESKTOP-I4F9RFA:~/w9/9cat-grep$ |
```

Figure 17

Step 8: Defining the alias again preserving it for the next session.

```

GNU nano 8.4 .bashrc *
PS1="${[\w]0;$(debian_chroot:+($debian_chroot))\u@h: \w\a}$PS1"
# enable color support of ls and also add handy aliases
if [ -x /usr/bin/dircolors ]; then
    test -r ~/.dircolors && eval "$(dircolors -b ~/.dircolors)" || eval "$(dircolors -b)"
    alias ls='ls --color=auto'
    #alias dir='dir --color=auto'
    #alias vdir='vdir --color=auto'

    #alias grep='grep --color=auto'
    #alias fgrep='fgrep --color=auto'
    #alias egrep='egrep --color=auto'
fi

# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

# some more ls aliases
#alias ll='ls -l'
#alias la='ls -A'
#alias l='ls -CF'

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
    if [ -x /usr/share/bash-completion/bash_completion ]; then
        . /usr/share/bash-completion/bash_completion
    elif [ -x /etc/bash-completion/bash_completion ]; then
        . /etc/bash-completion
    fi
fi
alias lsal='ls -al'
Write to File: .bashrc
Help Cancel DOS Format Mac Format Append Prepend Backup File Browse

```

Figure 18

```

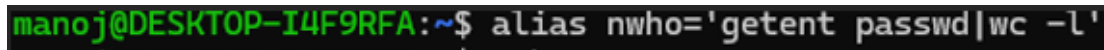
manoj@DESKTOP-I4F9RFA: ~
manoj@DESKTOP-I4F9RFA:~$ nano .bashrc
manoj@DESKTOP-I4F9RFA:~$ alias
alias ls='ls --color=auto'
alias lsal='ls -al'
manoj@DESKTOP-I4F9RFA:~$ |

```

Figure 19

Step 9: Define the nwho alias for the number of system file at UNIX computers.

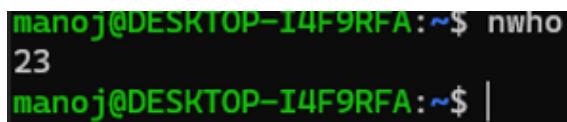
alias nwho='getent passwd|wc -l'



```
manoj@DESKTOP-I4F9RFA:~$ alias nwho='getent passwd|wc -l'
```

Figure 20

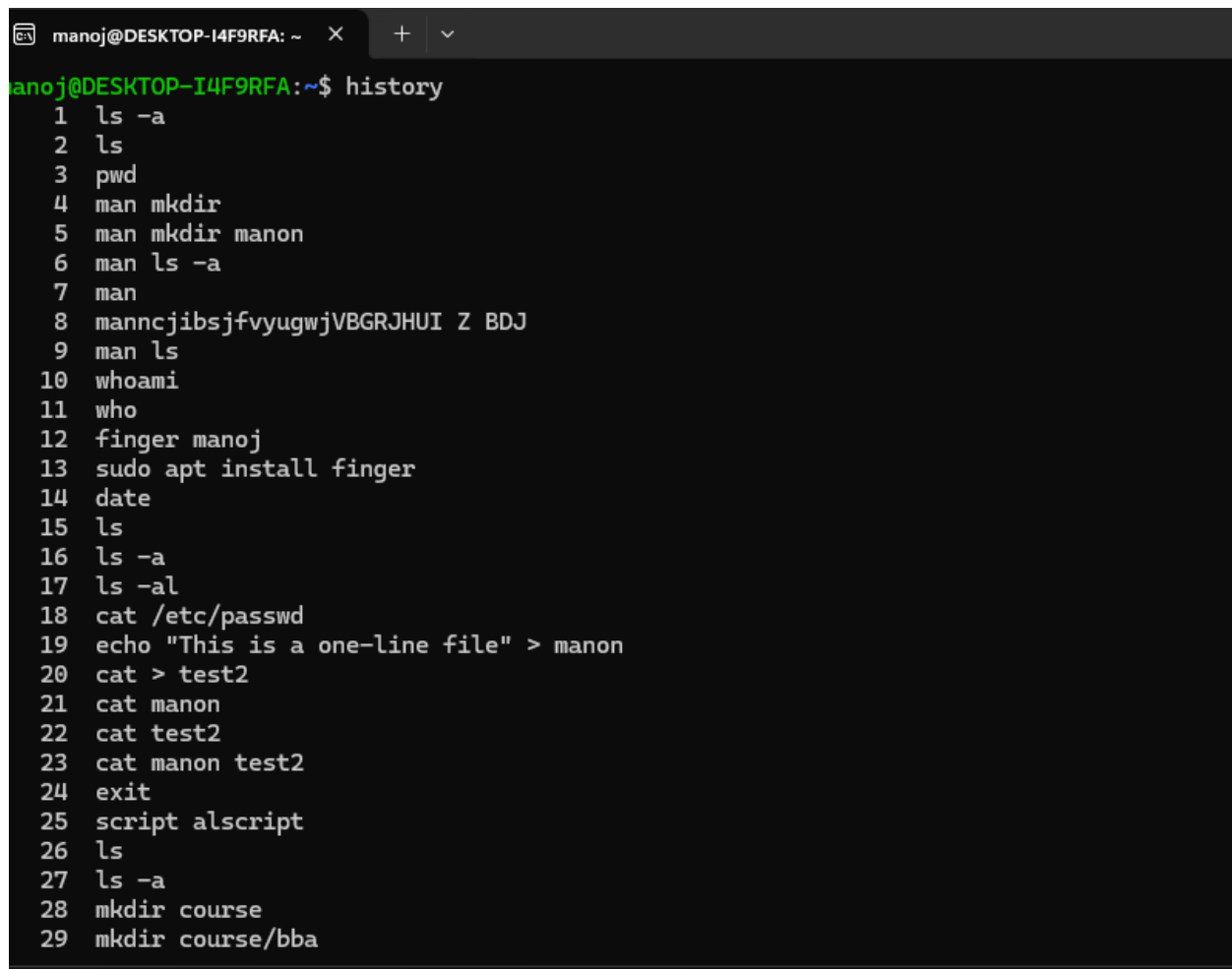
Step 10: Give the command nwho. Compare the figure displayed with ones got by your UNIX-mates.



```
manoj@DESKTOP-I4F9RFA:~$ nwho
23
manoj@DESKTOP-I4F9RFA:~$ |
```

Figure 21

Step 11: List your last commands executed giving the history command.

A terminal window with a dark background and light green text. The window title bar shows 'manoj@DESKTOP-I4F9RFA: ~' and standard window controls. The terminal displays the command 'history' and its output, which is a numbered list of 29 commands executed in the session.

```
manoj@DESKTOP-I4F9RFA:~$ history
1  ls -a
2  ls
3  pwd
4  man mkdir
5  man mkdir manon
6  man ls -a
7  man
8  manncjibsjfvyugwjVBGRJHUI Z BDJ
9  man ls
10 whoami
11 who
12 finger manoj
13 sudo apt install finger
14 date
15 ls
16 ls -a
17 ls -al
18 cat /etc/passwd
19 echo "This is a one-line file" > manon
20 cat > test2
21 cat manon
22 cat test2
23 cat manon test2
24 exit
25 script alscript
26 ls
27 ls -a
28 mkdir course
29 mkdir course/bba
```

Figure 22



```
manoj@DESKTOP-I4F9RFA: ~  
283 grep -n ll testa  
284 grep -l ll*  
285 grep -l ll *  
286 grep -i ll *  
287 grep -i LL  
288 grep -i LL *  
289 grep -c ll *  
290 grep '^K' testa testb  
291 grep -n '^' testa  
292 grep -n '^' testa  
293 alias lsal = 'ls -al'  
294 alias lsal='ls -al'  
295 lsal  
296 unalias lsal  
297 lsal  
298 pwd  
299 nano .bashrc  
300 alias  
301 clear  
302 nano .bashrc  
303 alias  
304 alias nwho='getent passwd|wc -l'  
305 nwho  
306 tree\  
307 cd w9/9cat-grep  
308 alias lsal='ls -al'  
309 alias  
310 alias .bashsc  
311 history  
manoj@DESKTOP-I4F9RFA:~$
```

Figure 23

Step 12: Re-execute the last but one command using the redo (r) command and the number of the event.fc -r



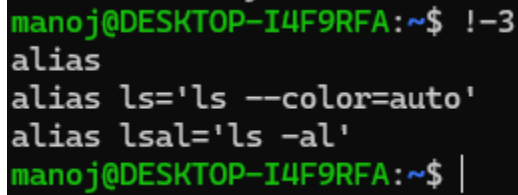
The screenshot shows a terminal window with the GNU nano 8.4 text editor. The window title is "manoj@DESKTOP-I4F9RFA: ~". The editor is editing a file named "/tmp/bash-fc.dKpGUh". The command history is visible, showing the command "event.fc -r" being entered. The nano editor's status bar at the bottom displays various keyboard shortcuts for editing and navigation.

```
manoj@DESKTOP-I4F9RFA: ~  
GNU nano 8.4 /tmp/bash-fc.dKpGUh  
history  
event.fc -r  
[ Read 1 line ]  
G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute  
X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify  
^C Location  M-U Undo     M-A Set Mark  
^_ Go To Line M-E Redo     M-6 Copy
```

Figure 24

Step 13: Re-execute the command given three commands ago using the negative integer.

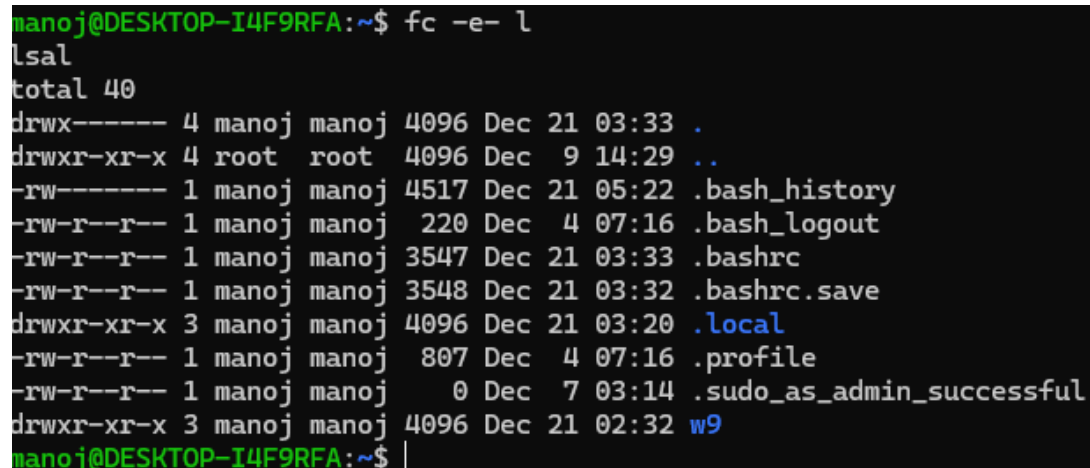
!-3



```
manoj@DESKTOP-I4F9RFA:~$ !-3
alias
alias ls='ls --color=auto'
alias lsal='ls -al'
manoj@DESKTOP-I4F9RFA:~$ |
```

Figure 25

Step-14: Re-execute the last command which name begins with 'l'.fc -e- l



```
manoj@DESKTOP-I4F9RFA:~$ fc -e- l
lsal
total 40
drwx----- 4 manoj manoj 4096 Dec 21 03:33 .
drwxr-xr-x 4 root root 4096 Dec 9 14:29 ..
-rw----- 1 manoj manoj 4517 Dec 21 05:22 .bash_history
-rw-r--r-- 1 manoj manoj 220 Dec 4 07:16 .bash_logout
-rw-r--r-- 1 manoj manoj 3547 Dec 21 03:33 .bashrc
-rw-r--r-- 1 manoj manoj 3548 Dec 21 03:32 .bashrc.save
drwxr-xr-x 3 manoj manoj 4096 Dec 21 03:20 .local
-rw-r--r-- 1 manoj manoj 807 Dec 4 07:16 .profile
-rw-r--r-- 1 manoj manoj 0 Dec 7 03:14 .sudo_as_admin_successful
drwxr-xr-x 3 manoj manoj 4096 Dec 21 02:32 w9
manoj@DESKTOP-I4F9RFA:~$ |
```

Figure 26

1.5 Conclusion

The lab provided hands-on experience with some essential UNIX utilities for file handling. The use of grep with various options helps us for powerful text searching capabilities. We have also learn a shortcut command in UNIX knows as alias. We can create shortcut command temporary and permanently. To keep it forever, we must add it to .bashrc hidden folder. Overall, this lab helps me understand how to search files and create shortcuts in the UNIX environment.